

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the batteries. Refer to the original equipment manufacturer (OEM) service manual.
- To measure the camshaft end clearance without removing the front gear cover, first remove the rocker lever covers. Refer to Procedure 003-011 in Section 3.
(</qs3/pubsys2/xml/en/procedures/57/57-003-011-tr.html>)

Loosen the locknut and turn out the adjusting screw on all of the valve rocker levers. The overhead load on the camshaft **must** be released to allow free movement of the camshaft.

Note : It is **not** necessary to remove the push rods.

Left Bank

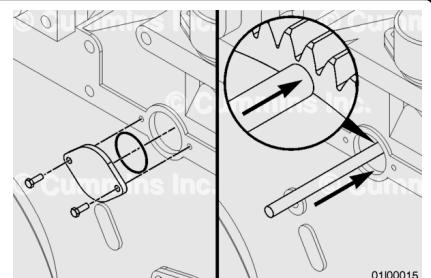
Remove the two capscrews and cover plate in the front of the gear cover. Discard the o-ring.

Use a rod against the camshaft gear to push the camshaft as far as possible toward the rear of the engine.

Remove the rod.

Remove the camshaft rear cover plate. Refer to Procedure 001-011 in Section 1.

(</qs3/pubsys2/xml/en/procedures/57/57-001-011-tr.html>)



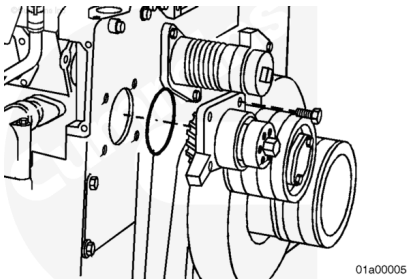
Right Bank

Remove the charging system alternator drive and charging system alternator drive pulley. Refer to Procedure 009-008 in Section 9. (/qs3/pubsys2/xml/en/procedures/57/57-009-008.html) Refer to Procedure 009-010 in Section 9. (/qs3/pubsys2/xml/en/procedures/57/57-009-010.html)

Use a rod against the camshaft gear to push the camshaft as far as possible toward the rear of the engine.

Remove the rod.

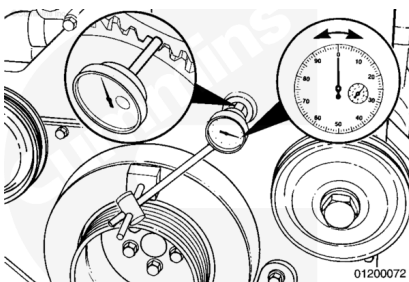
Remove the camshaft rear cover plate. Refer to Procedure 001-011 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-011-tr.html)



Measure

Set up a dial indicator gauge, Part Number 3376050, and extension, Part Number ST-537-4, with a magnetic base on the front of the cylinder head or vibration damper. Place the tip of the indicator on the face of the camshaft gear through the gear cover inspection hole.

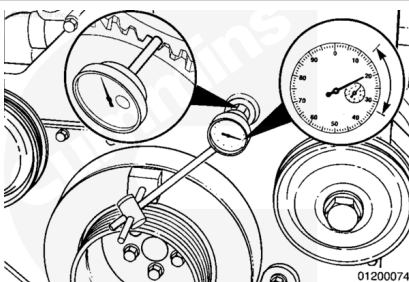
Set the dial indicator to zero.



Apply pressure forward on the camshaft from the rear access where the cover plate was removed. Record the measurement on the dial indicator.

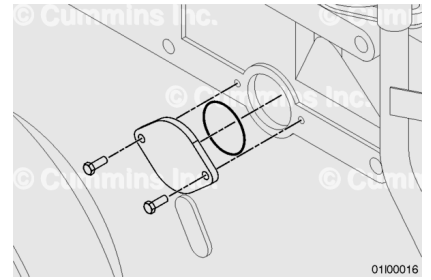
Camshaft End Clearance		
mm		in
0.100	MIN	0.004
0.280	MAX	0.011

If the camshaft end clearance does **not** meet the above specifications, the front cover **must** be removed and the camshaft thrust plate inspected. Refer to Procedure 001-008 in Section 1 (/qs3/pubsys2/xml/en/procedures/57/57-001-008-tr.html).



Install the two capscrews and cover plate in the front of the gear cover with a new o-ring.

Torque Value: 14 n·m [124 in-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the charging system alternator drive. Refer to Procedure 009-008 in Section 9. (</qs3/pubsys2/xml/en/procedures/57/57-009-008.html>)
- Install the charging system alternator drive pulley. Refer to Procedure 009-010 in Section 9. (</qs3/pubsys2/xml/en/procedures/57/57-009-010.html>)
- Adjust the valves. Refer to Procedure 003-004 in Section 3 (</qs3/pubsys2/xml/en/procedures/57/57-003-004-tr.html>).
- Install the rocker lever covers. Refer to Procedure 003-011 in Section 3. (</qs3/pubsys2/xml/en/procedures/57/57-003-011-tr.html>)
- Install the camshaft rear cover plate. Refer to Procedure 001-011 in Section 1 (</qs3/pubsys2/xml/en/procedures/57/57-001-011-tr.html>).
- Connect the batteries. Refer to the OEM service manual.
- Start the engine and check for proper operation.

